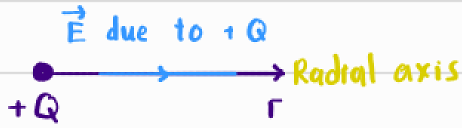
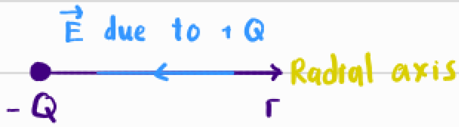
E-field [Coulomb's Law], E

- Electric force per unit positive charge





* Important terms

- Point charge [Source Charge], Q
  - Any charge, can't move, no volume
- Test charge, q
  - Any charge, can move, have volume

$$|\vec{E}| = \frac{F_E}{q} = \left[\frac{kQq}{r^2}\right]\left[\frac{1}{q}\right]$$

$$\boxed{|\vec{E}| = \frac{kQ}{r^2}}, \text{ where } k = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9$$

Permitivity of free space [8.85×10^{-12}]

Electric Flux [Gauss's Law], ϕ_E

Doesn't have to be a sphe

- Total electric flux in a closed surface is proportional to the total charge enclosed within that surface.

Gaussian Surfc

Integral over closed surface

Normal vec pointing out from the enclosing surface

$$\boxed{\text{Electric Flux } \phi_E = \oint \vec{E} \cdot \vec{dA}}$$

$$\phi_E = \vec{E} \oint \vec{dA}$$

Sum of tiny areas

Doesn't have to be a sphere to proof the formula

$$= \left[\left(\frac{1}{4\pi\varepsilon_0}\right)\left(\frac{Q}{r^2}\right)\right]\left[4\pi r^2\right]$$

[Assuming the closed surface is a sphere]

Gauss's Law : $\boxed{\phi_E = \frac{Q_{enclosed}}{\varepsilon_0}}$

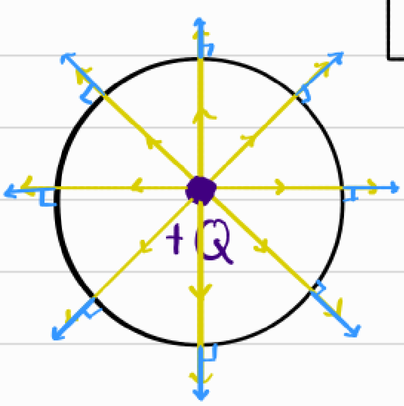
* Note :

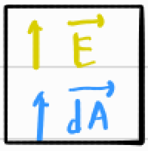
$\phi_E = \vec{E} \cdot \vec{dA}$

$\phi_E = E\,dA \cos\theta$

$\therefore \theta = 0°, \phi_E = \max$

$\theta = 90°, \phi_E = 0$





Gauss's Law

By convention, stronger E = more lines per volume

By knowing the number of lines going through any closed surface we can know the $Q_{enclosed}$.

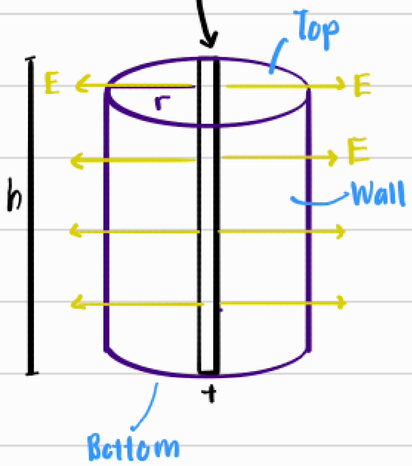
For a specific Q,

"lines density ↓" "More area to cut the lines"

If $r\uparrow$, $E\downarrow$ but $A\uparrow$, so no. of lines going through the closed surface is constant, vice versa.

Line of charge

Line of charge [Assume Charge density, λ is constant throughout the rod]



Total Electric Flux $\Sigma\phi_E$

$$\Sigma\phi_E = \phi_{E\,top} + \phi_{E\,bottom} + \phi_{E\,wall}$$

$$\frac{Q_{enclosed}}{\varepsilon_0} = 0 + 0 + E(dA)\cos 0°$$

$$\boxed{Q_{enclosed} = (E)(dA)(\varepsilon_0)}$$

$$\lambda h = E(2\pi r h)\varepsilon_0$$

$$\boxed{E = \frac{\lambda}{(2\pi r)\varepsilon_0}}$$ Charge density

Electric Force, F_E

- Actual ammount of force exerted on a positive test charge, q by an electric field

$$E = \frac{F_E}{q}$$ [From E-field]

$$F_E = qE$$

$$\boxed{F_E = \frac{kQq}{r^2}}$$

Conductors in Electrostatic Equilibrium

- Electrostatic Equilibrium
  - ↳ Charges are at rest
    - ↳ Net $F_E = 0$
      - ↳ $E = 0$

Assumption: All conductors are in Electro. equi.

$\therefore$ $E = 0$ inside the conductor. So if the conductor has a net charge, the charge is definately on the surface of the conductor.

[Excess charge accumulate & distribute itself on the outer surface of the conductor]